



SIMATSENGINEERING

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Course Code: CSA1712

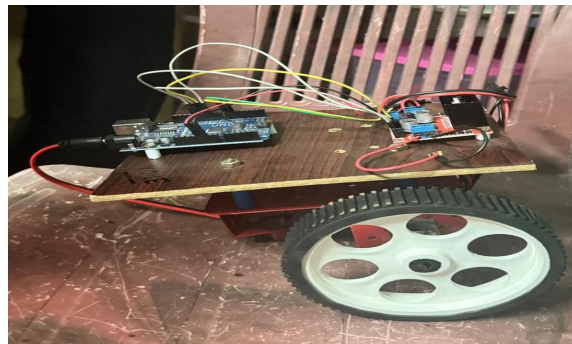
Slot: C

Course Name: Artificial Intelligence

Course Faculty: Dr. R. Veeramani

Project Title: “Smart Robot Car Using Arduino and L298N Motor Driver”

MODULE 3: MOTOR CONTROL DRIVER – L298N



Project Description:

The Smart Robot Car Using Arduino and L298N Motor Driver is an AI-assisted robotic vehicle developed to provide intelligent, wireless, and user-friendly control of a mobile robot. The system integrates an Arduino UNO, HC-05 Bluetooth module, L298N motor driver, and DC motors to achieve reliable movement and communication. A mobile phone is used to send movement commands wirelessly through the HC-05 Bluetooth module to the Arduino UNO, which processes the received commands and generates appropriate control signals for the L298N motor driver. The motor driver regulates the direction and movement of the connected DC motors, allowing the robot to move according to the user's instructions. Artificial Intelligence is incorporated into the system to support intelligent decision-making and enable the robot to respond more effectively to its operating environment, providing a foundation for future autonomous features such as obstacle detection, intelligent navigation, and automated movement. The project combines AI, embedded systems, microcontroller programming, wireless communication, motor control, and robotics to demonstrate how intelligent technologies can be integrated with hardware to develop practical robotic applications. The system also provides a flexible platform that can be enhanced with additional sensors, IoT connectivity, mobile applications, and advanced AI algorithms for improved automation and autonomous operation.

Student Signature

Guide Signature